# serviceOrchestrationSimpleStoreManagement

## Service Description

Service ID: *"serviceOrchestrationSimpleStoreManagement"*

**Abstract**
This document describes the `serviceOrchestrationSimpleStoreManagement` service, version 1.0.0, produced by the SimpleStoreServiceOrchestration core system of Arrowhead 5.2.2.

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# 1 Overview

This document describes the `serviceOrchestrationSimpleStoreManagement` service, which provides operator management of the orchestration store: the static rules telling which service instance a consumer shall use, with priorities. It is produced by the SimpleStoreServiceOrchestration system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 2, we describe the abstract message operations provided by the service. In Section 3, we end the document by presenting the data types used by the mentioned operations. The concrete interfaces are specified in the HTTP/TLS/JSON IDD [2] and the MQTT/TLS/JSON IDD [3].

## 1.1 Significant Prior Art

Successor of the Arrowhead 4.x Orchestrator store management.

## 1.2 How This Service Is Meant to Be Used

Operators and support systems such as the PlantDescriptionEngine create rules (consumer, service instance, priority), query, reprioritize and remove them. `pull` then returns the rules of the consumer in priority order.

## 1.3 Important Delimitations and Dependencies

Rules are not checked against the ServiceRegistry or ConsumerAuthorization at orchestration time.

# 2 Service Interface

This section describes the interfaces to the `serviceOrchestrationSimpleStoreManagement` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 3.




Figure 1: SysML-style block of the `serviceOrchestrationSimpleStoreManagement` service and its operations.

The following interface operations are available.

## 2.1 operation query (OrchestrationSimpleStoreQueryRequest) : OrchestrationSimpleStoreListResponse

Returns a page of store entries matching the filter.

## 2.2 operation remove (List<String>)

Removes store entries by identifier.

## 2.3 operation create (OrchestrationSimpleStoreListRequest) : OrchestrationSimpleStoreListResponse

Creates store entries; the service definition is taken from the instance identifier.

## 2.4 operation modify-priorities (PriorityRequest) : OrchestrationSimpleStoreListResponse

Changes the priorities of store entries.

# 3 Information Model

Here, all data objects that can be part of the `serviceOrchestrationSimpleStoreManagement` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 3.8, which are used to represent things like hashes and identifiers.

## 3.1 struct OrchestrationSimpleStoreListRequest

Store entries to create.

| Field | Type | Description |
|---|---|---|
| candidates | List<OrchestrationSimpleStoreRequest> | Store entries to create. |

## 3.2 struct OrchestrationSimpleStoreListResponse

List of store entries.

| Field | Type | Description |
|---|---|---|
| entries | List<OrchestrationSimpleStoreResponse> | Matching records of the current page. |
| count | Number | Total number of matching records (all pages). |

## 3.3 struct OrchestrationSimpleStoreQueryRequest

Paged filter for store entries.

| Field | Type | Description |
|---|---|---|
| pagination | Page | Paging and sorting of the result. |
| ids | List<Identifier> | Store entry identifiers (UUID) to match. |
| consumerNames | List<String> | Consumer system names to match. |
| serviceDefinitions | List<Name> | Service definition names to match. |
| serviceInstanceIds | List<Identifier> | Service instance identifiers to match. |
| minPriority | Number | Lower bound of the priority. |
| maxPriority | Number | Upper bound of the priority. |
| createdBy | Name | Creator system to match. |

## 3.4 struct OrchestrationSimpleStoreRequest

One orchestration rule: consumer, service instance and priority.

| Field | Type | Description |
|---|---|---|
| consumer | Name | Consumer system the rule applies to. |
| serviceInstanceId | Identifier | Service instance offered to the consumer. |
| priority | Number | Priority of the rule; 1 is the highest. |

## 3.5 struct OrchestrationSimpleStoreResponse

A stored orchestration rule.

| Field | Type | Description |
|---|---|---|
| id | Identifier | Identifier (UUID) of the store entry. |
| consumer | Name | Name of the consumer system. |
| serviceDefinition | Name | Service definition name. |
| serviceInstanceId | Identifier | Service instance identifier `<System>\|<serviceDefinition>\|<version>`. |
| priority | Number | Priority of the entry; 1 is the highest. |
| createdBy | Name | Name of the system that created the record. |
| updatedBy | Name | Name of the system that changed the record last. |
| createdAt | DateTime | UTC time of creation. |
| updatedAt | DateTime | UTC time of the last change. |

## 3.6 struct Page

Paging and sorting; all fields optional.

| Field | Type | Description |
|---|---|---|
| page | Number | Zero-based page number. |
| size | Number | Number of records per page (at most `max.page.size`, default 1000). |
| direction | String | Sort direction, `ASC` or `DESC`. |
| sortField | String | Field to sort by. |

## 3.7 struct PriorityRequest

A JSON object mapping store entry identifiers (UUID) to new priorities, e.g. `{"123e4567-e89b-12d3-a456-426614174000": 5}`.

## 3.8 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

| Type | Description |
|---|---|
| DateTime | Pinpoints a specific moment in time (UTC). |
| Identifier | A string that identifies a record, e.g. a UUID or a composite identifier. |
| List | An *array* of a known number of items, each having type A. |
| Name | A string identifier that is intended to be both human and machine-readable. |
| Number | An integer number. |
| String | A chain of characters. |

# 4 References

[1] J. Delsing, “SimpleStoreServiceOrchestration – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, serviceorchestration-simple/documentation/v5.2.2/SimpleStoreServiceOrchestration-v5.2.2-SysD.pdf.

[2] ——, “serviceOrchestrationSimpleStoreManagement HTTP/TLS/JSON – Interface Design Description (SimpleStoreServiceOrchestration),” Eclipse Arrowhead, IDD 5.2.2, 2026, simpleStoreServiceOrchestration-v5.2.2-serviceOrchestrationSimpleStoreManagement-HTTP-IDD.pdf.

[3] ——, “serviceOrchestrationSimpleStoreManagement MQTT/TLS/JSON – Interface Design Description (SimpleStoreServiceOrchestration),” Eclipse Arrowhead, IDD 5.2.2, 2026, simpleStoreServiceOrchestration-v5.2.2-serviceOrchestrationSimpleStoreManagement-MQTT-IDD.pdf.

# 5 Revision History

## 5.1 Amendments

| No. | Date | Version | Subject of Amendments | Author |
|---|---|---|---|---|
| 1 | 2026-10-22 | 5.2.2 | Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace. | Jerker Delsing |

## 5.2 Quality Assurance

| No. | Date | Version | Approved by |
|---|---|---|---|
| 1 | | 5.2.2 | |